

Day 8 – Cyber Security & Hacking

Submarine Cables: The Hidden Physical Internet

- The internet is **not fully wireless**
- Most global data does **not travel via satellites**
- **Over 95% of the world's internet traffic** moves through **submarine cables**
- These cables run **under oceans and seas**, connecting continents



♦ Definition: What Are Submarine Cables?

- Submarine cables are **fiber-optic cables laid on the ocean floor**
- They physically connect:
 - Countries
 - Continents
 - Data centers
- They transmit data using **light signals** at extremely high speed

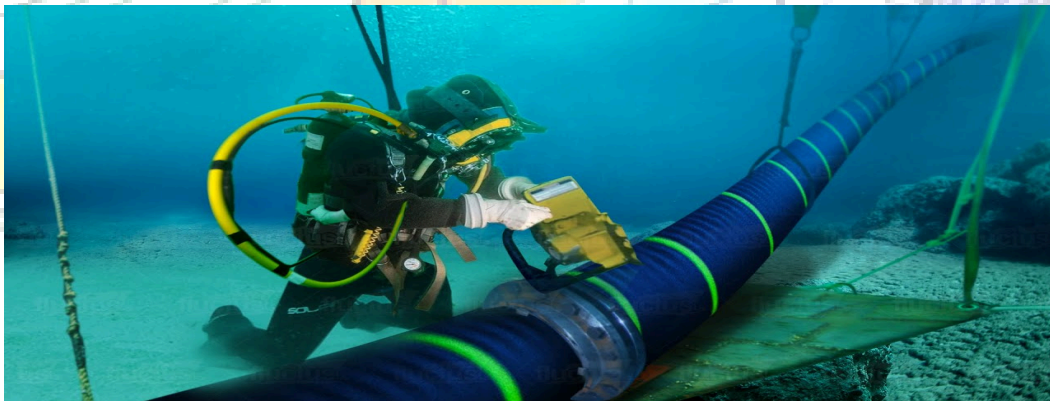


♦ How Internet Data Physically Travels

- User sends data (email, website request, payment)
- Data is converted into **light pulses**
- Light travels through **glass fibers** inside submarine cables
- Signals pass through:
 - Cable landing stations
 - Network exchange points
- Data reaches another continent in milliseconds

♦ Why This Is Important for Cyber Security

- Cyber systems depend on **physical infrastructure**
- If cables are damaged:
 - Internet slows or stops
 - Banking systems fail
 - Cloud services go offline
- Cyber attacks can exploit **physical weaknesses**



♦ Cyber + Physical Attack Connection

- Attackers may:
 - Target cable landing stations
 - Exploit reduced connectivity during outages
 - Launch DDoS attacks when redundancy is low
- This creates **hybrid attacks** (physical + cyber)